use statistics nationwide in hundreds of cities and municipalities at a much faster pace.

Widespread embedded safety alert systems. This approach allows *ad-hoc* technology disruptors and creative individuals to embed these sensors more broadly in smaller and safer stationary use cases, creating smart networks of lidar and embedded sensors.

A public safety system like this might provide a new kind of stationary collision-avoidance and weather systems for congested roads, toll ways, on-ramps/off-ramps, alerts for drivers about pedestrians, school crossings, bikes, narrowed lanes, accident avoidance, construction and wildlife. By sensing the automobile behaviour, it could alert drivers of roadway weather conditions (black ice, ice on bridges, snow conditions, slick roads, etc). All modes of traffic and weather conditions could be understood more deeply. In this case, any new technology ideas in these application areas would be better than none—the situation we have today.

These sensor systems are continuing to get smaller and will be ideal for a number of embedded applications. Instruments and techniques such as the compass, sextant, LORAN radiolocation and dead reckoning. These are among those that have been used with varying degrees of accuracy, consistency and availability.

For autonomous vehicles, the navigation and guidance sub-system must always be active and keep checking how the vehicles are doing versus the goal. For example, if the originally optimum route has unexpected diversions, the path must be re-computed in real time to avoid going in the wrong direction. Since the vehicles are obviously constrained to roadways, this takes much more computational effort than simply drawing a straight line between A and B.

The primary sub-system used for navigation and guidance is based on a GPS receiver, which computes the present position based on complex analysis of signals received from at least four constellations of over 60 low-orbit satellites. A GPS system can provide location accuracy of the order of one metre (actual number depends on many subtle issues), which is a good start for the vehicle. Note that, for a driver, who hopes to hop in the car and get going, a GPS receiver takes between 30 and 60 seconds to establish initial position, so the autonomous vehicle must delay its departure until this first fix is computed.

GPS sub-systems are now available as sophisticated system-on-chip (SoC) ICs or multi-chip chipsets that require only power and antenna, and include an embedded, application-specific compute engine to perform intensive calculations. Although many of these ICs have an internal RF preamp for the 1.5GHz GPS signal, many vehicles opt to put the antenna on the roof with a co-located low-noise amplifier (LNA) or RF preamplifier, and locate the GPS circuitry in a more convenient location within the vehicle. The antenna must have right-hand circular polarisation characteristics to match the polarisation of GPS signals, and it can be a ceramic-chip unit, a small wound stub design or have a different configuration.

Fig. 4: A complete GPS module

An example of a GPS module is RXM-GPS-F4-T from Linx Technologies, shown in Fig. 4. This 18mm × 13mm × 2.2mm surface-mount unit requires a single 1.8V supply at 33mA, and can acquire and track up to 48 satellites simultaneously—more channels allow the GPS to see and capture more data and, thus, yield better results and fewer drop-